Farhan Mahmood

Software Developer | Node.js | Fullstack Development | REST APIs | MongoDB | React.js | AI/LLM Integration

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EXPERIENCE

Full-Stack Developer Apr 2023 – Present
Grubweb Service

- Designed and developed an internal **Project & Client Management System** covering client and project management, employee assignments, task tracking, and the complete project lifecycle from onboarding to delivery.
- Built and shipped end-to-end web products for clients across **real estate, healthcare, e-commerce, and consultancy** using React.js, Next.js, Node.js, and TypeScript.
- Developed **server-side rendered applications** using Next.js App Router, focusing on fast load times and strong Core Web Vitals.
- Built RESTful APIs with **JWT authentication and role-based access control (RBAC)** for Admin, Manager, and Developer roles.
- Designed a normalized **PostgreSQL database** using Prisma ORM for clients, projects, tasks, feedback, and employee management.

PROJECTS

Enterprise AI Knowledge & Workflow Copilot(Building)
Java 21 | Spring Boot | React | Python | FastAPI | PostgreSQL | pgvector | RAG | LLMs | Docker
https://github.com/farhan2804/enterprise-ai-copilot

- Designed an AI-powered enterprise knowledge assistant using a Retrieval-Augmented Generation approach, an LLM, and semantic search to retrieve relevant information from enterprise documents and provide contextually grounded answers with sources.
- Engineered a document ingestion/retrieval pipeline involving text extraction, chunking, embedding, similarity search, and metadata filtering using PostgreSQL and pgvector.
- Developed the full-stack application with React, designed the REST APIs with Spring Boot, built the Python/FastAPI backend with AI services, set up the auth system and role-based access management, and deployed the application with Docker. Additionally, planned to evaluate the RAG results, measure the retrieved information quality, and automate the AI processes.

Diagnostic Test Booking & Payment Management System
Java 21 | Spring Boot | Spring Security | JWT | PostgreSQL | Spring Data JPA | REST APIs | Swagger/OpenAPI | JUnit/Mockito
https://github.com/farhan2804/eve_healthcare_booking_api

- Developed a backend healthcare platform supporting **authentication, diagnostic test management, appointment booking and payment workflows**.
- Designed REST APIs and relational database schemas for users, diagnostic centres, tests, bookings and payments.
- Implemented request validation, authorization, payment ownership checks, error handling and **idempotent payment webhooks**.
- Added Swagger/OpenAPI documentation, structured logging, JUnit/Mockito unit tests and integration tests, with **9 automated tests passing**.

AI SQL Assistant – Natural Language to SQL Engine
Python/LLM Integration | Groq Llama 3.3 | Java | Spring Boot | React.js | PostgreSQL | REST APIs
https://aisqlassistant.netlify.app/

- Developed a full-stack application that converts **natural-language requests into SQL queries** using an LLM API.
- Built Spring Boot REST APIs for SQL generation and PostgreSQL database interactions.
- Developed React components for query input, API communication and displaying generated SQL and results.
- Deployed the application using **Netlify, Railway and Neon PostgreSQL**.

SKILLS

Languages: Java, JavaScript

Frontend: React.js, Next.js, HTML5, CSS3, Responsive Web Design

Backend: Node.js, Express.js, REST APIs, Server-Side Business Logic, API Integration

Databases: PostgreSQL, MySQL, MongoDB

ORM & Data Access: Spring Data JPA

Authentication & Security: JWT Authentication, Role-Based Access Control (RBAC)

AI & LLM: LLM API Integration, Groq API, Prompt Engineering Fundamentals

Tools & Development Practices: Git, GitHub, Postman, Swagger/OpenAPI, Debugging, API Testing

Core Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Database Design, Error Handling, Software Development Lifecycle (SDLC)

EDUCATION

Bachelor of Technology (B.Tech) Aug 2021 – Jun 2024
Electronics and Communication Engineering

University: Jharkhand University of Technology
CGPA: 8.04

CERTIFICATIONS

Microsoft Certified: Azure Cloud Fundamentals (AZ-900)

Microsoft Certified: Azure AI Fundamentals (AI-900)

ADDITIONAL — TECHNICAL EDUCATOR, YOUTUBE

Engineering & Web Development Channel
https://www.youtube.com/@formulaC_Coder2804